

# **AN EXPLAINABLE FEATURE-BASED SMARTPHONE RECOMMENDATION SYSTEM USING ASPECT-BASED SENTIMENT ANALYSIS (ABSA) OF USER REVIEWS**

Submitted by

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## Declaration

I declare that this dissertation is my own work and has not been submitted, either in whole or in part, for any other degree or academic qualification. All sources used in developing the study have been acknowledged through appropriate citations and references.

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Date: \_\_\_\_13/07/2026\_\_\_\_

## Abstract

Smartphone purchasers have to compare many products and interpret huge amounts of online reviews before deciding to purchase. Traditional recommendation methods typically rely on aggregate ratings or past user behaviors, which may obscure feature-level pros and cons and offer limited explanation for why a particular smartphone is recommended. This paper proposes an explainable feature-based smartphone recommendation system that utilizes Aspect-Based Sentiment Analysis (ABSA) to transform Amazon user reviews into structured product evidence.

The research is based on a pragmatic philosophy and a primarily quantitative, computational approach within an applied design-and-development strategy. Publicly available Amazon smartphone reviews in English will be downloaded and stored in a static offline dataset. The dataset will be cleaned, tokenized and analyzed for six pre-defined features: camera, battery, display, performance, design and price. For each identified feature, a hybrid aspect identification approach and a transformer based sentiment classifier will be used to determine whether the sentiment is positive, negative or neutral. The resulting aspect-level outputs are aggregated into feature performance scores and combined with user-defined feature weights to rank smartphones according to individual priorities.

The recommendation interface will provide evidence-based explanations based on the same feature scores, weighted contributions and review counts as the ranking process. The technical evaluation will measure accuracy, precision, recall, F1-score for aspect identification and sentiment classification, and end-to-end ABSA performance. Moreover, the feature-based ranking and explanation output will be compared against a traditional average-rating baseline by means of structured user-centered evaluation. We expect the study to result in an integrated and transparent framework linking review collection, ABSA, feature scoring, user-priority weighting, recommendation ranking and explanation generation in a single smartphone-specific decision-support prototype.

**Keywords:** Aspect-Based Sentiment Analysis; smartphone recommendation; explainable artificial intelligence; Amazon reviews; feature-based recommendation; sentiment-aware recommendation.

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## Acronyms

| Acronym       | Meaning                                                 |
|---------------|---------------------------------------------------------|
| <b>ABSA</b>   | Aspect-Based Sentiment Analysis                         |
| <b>AI</b>     | Artificial Intelligence                                 |
| <b>API</b>    | Application Programming Interface                       |
| <b>ASIN</b>   | Amazon Standard Identification Number                   |
| <b>BERT</b>   | Bidirectional Encoder Representations from Transformers |
| <b>BiLSTM</b> | Bidirectional Long Short-Term Memory                    |
| <b>CNN</b>    | Convolutional Neural Network                            |
| <b>CSV</b>    | Comma-Separated Values                                  |
| <b>HTML</b>   | HyperText Markup Language                               |
| <b>JSON</b>   | JavaScript Object Notation                              |
| <b>NDCG</b>   | Normalized Discounted Cumulative Gain                   |
| <b>NLP</b>    | Natural Language Processing                             |
| <b>SHAP</b>   | SHapley Additive exPlanations                           |
| <b>SVM</b>    | Support Vector Machine                                  |
| <b>TF-IDF</b> | Term Frequency-Inverse Document Frequency               |
| <b>XAI</b>    | Explainable Artificial Intelligence                     |

# Chapter 1: Introduction

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## 1.1 Chapter Introduction

The growth of e-commerce has expanded smartphone choice, but it has also increased the effort needed to compare models and interpret large volumes of reviews. Buyers usually evaluate several attributes-camera, battery, display, performance, design, and price while conventional systems often reduce these experiences to overall ratings or interaction patterns. Such outputs may identify popular products without clearly showing whether they match an individual user's priorities.

This study proposes an explainable feature-based smartphone recommendation system using Aspect-Based Sentiment Analysis (ABSA) of Amazon user reviews. The system extracts feature level sentiment, creates product feature scores, applies user-defined weights, ranks smartphones, and explains the result in plain language. This chapter presents the research background, problem, aim, approved objectives and questions, significance, scope, expected outcomes, and dissertation structure.

## 1.2 Background of the Study

### 1.2.1 Recommendation Systems in E-commerce

E-commerce recommendation systems commonly use collaborative filtering, content-based filtering, or hybrid models. These approaches can reduce search effort and model complex user-item relationships, but collaborative methods depend on sufficient historical interactions and may perform poorly for new users or products (Sami et al., 2024; Volkovs et al., 2017; Lin, 2022).

A second limitation is weak transparency. Smartphone buyers make different feature trade-offs, yet an overall score rarely shows whether a recommendation is driven by camera, battery, price, or another attribute. Aspect- and review-aware studies indicate that review sentiment can be represented as separate recommendation criteria, providing a clearer basis for ranking (Musto et al., 2017; Dang et al., 2021; Liu et al., 2021; Haghighi & Moradian Zadeh, 2025).

### **1.2.2 Smartphone Selection as a Multi-Feature Decision**

Smartphone selection is a multi-criteria decision involving technical specifications and real-use experience. Specifications describe components such as processors, memory, battery capacity, and camera resolution, but they do not fully reflect battery endurance, low-light image quality, heating, responsiveness, build quality, or value in daily use.

Because users assign different importance to camera, battery, display, performance, design, and price, the same phone may be appropriate for one buyer and unsuitable for another. Treating these features as separate dimensions makes trade-offs visible and allows the final recommendation to be traced to feature scores and explicit user priorities (Musto et al., 2017; Dietz et al., 2022).

### **1.2.3 Role of Online User Reviews**

Online reviews provide practical evidence about product use that is often absent from manufacturer descriptions. Smartphone reviewers discuss battery drain, camera quality, display brightness, gaming performance, heating, design, and value for money. However, the text is unstructured and may contain spelling errors, informal language, duplicate content, and several opinions in one sentence.

Research shows that mobile-phone and e-commerce reviews can be collected and analysed at aspect level, with transformer models performing strongly in contextual review language (Soni et al., 2024; Singhal et al., 2025; Rasappan et al., 2024). The study therefore uses a cleaned offline Amazon dataset and applies duplicate, relevance, and traceability controls because manipulated or repeated reviews can distort product scores (Jin et al., 2026).

### **1.2.4 Aspect-Based Sentiment Analysis**

ABSA identifies the feature discussed in a text and the sentiment directed toward it. Unlike document-level classification, it can preserve different opinions within the same review; for example, ‘the camera is excellent, but the battery drains quickly’ produces camera-positive and battery-negative records.

Recent ABSA research combines contextual transformer representations with linguistic, graph, sequential, or extraction components to improve aspect detection and sentiment classification (Aziz et al., 2024; Jayakody et al., 2024). Mobile-phone evidence also

supports fine-tuned BERT models for technical review language (Singhal et al., 2025). Accordingly, the proposed system uses a transformer-based sentiment component together with a transparent, predefined aspect scheme.

### **1.2.5 Feature Scoring and Explainable Recommendation**

The classified aspect records are aggregated into separate product scores for camera, battery, display, performance, design, and price. Users assign importance weights to these features, and the recommendation engine combines the weighted values to rank eligible smartphones. This approach supports personalization without requiring a previous interaction history (Musto et al., 2017; Haghighi & Moradian Zadeh, 2025).

Explanation is generated from the same evidence used for ranking. The interface can show which highly weighted features increased a phone’s score and identify important trade-offs. This links analytical outputs to understandable reasons and avoids presenting an unexplained recommendation (Mansilla et al., 2025; Chen et al., 2026; Zhu, 2025).

## **1.3 Research Problem**

Current smartphone recommendation methods commonly rely on overall ratings, historical interactions, or similarity patterns. Although useful for general ranking, they may conceal feature-level strengths and weaknesses. A highly rated phone can still receive repeated criticism for battery or camera performance, while a moderately rated device may perform strongly in the feature a particular buyer values most.

The reviewed literature confirms that ABSA can extract feature sentiment and that review-derived sentiment can support recommendation (Soni et al., 2024; Dang et al., 2021; Liu et al., 2021). Nevertheless, ABSA studies often end with classification metrics, while recommender studies frequently emphasize rating prediction or latent user-item modelling. Limited work integrates smartphone review collection, explicit feature scoring, user-defined weights, transparent ranking, and plain-language explanations in one system.

The research problem is therefore the insufficient transformation of smartphone review text into feature-level, user-priority-based, and explainable recommendations. This study addresses that problem through an end-to-end Amazon review, ABSA, scoring, ranking, and explanation framework.

## 1.4 Research Aim

The aim of this research is:

**To develop an explainable feature-based smartphone recommendation system using Aspect-Based Sentiment Analysis (ABSA) of user reviews.**

The system will analyze textual reviews, extract feature-level sentiments, compute feature performance scores, and generate transparent, requirement-driven recommendations without relying on user history.

## 1.5 Research Objectives

The following goals are established in order to fulfill the research aim:

1. To collect and prepare a structured dataset of smartphone reviews from Amazon for natural language processing and analysis
2. To identify predefined smartphone features from review text and classify the sentiment associated with each feature as positive, negative, or neutral using Aspect-Based Sentiment Analysis (ABSA).
3. To determine each smartphone's feature performance score by calculating the percentage of positive sentiment mentions for each feature.
4. To develop a recommendation engine that ranks smartphones using weighted feature scores based on user-defined feature preferences.
5. To produce AI explanation messages in plain language that support each suggestion using the actual feature scores.

## 1.6 Research Questions

This study seeks to answer the following research questions:

- 1. How can sentiment at the feature level be reliably extracted from reviews of smartphones?**
- 2. How can sentiment scores for features be combined to create useful performance metrics?**
- 3. How can explainable recommendations be produced using feature scores?**

#### 4. When compared to conventional rating-based systems, does feature-based recommendation enhance interpretability

Table 1.1: Alignment of Research Questions and Research Objectives

| <b>Research Question</b>                                        | <b>Related Objective(s)</b>           | <b>Purpose</b>                                                                                                                                           |
|-----------------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| RQ1: Feature-level sentiment extraction                         | Objectives 1 and 2                    | To prepare the review dataset, identify the predefined smartphone features, and classify sentiment for each detected feature.                            |
| RQ2: Feature performance measurement                            | Objective 3                           | To convert classified feature sentiments into measurable performance scores for each smartphone.                                                         |
| RQ3: Explainable recommendation generation                      | Objectives 4 and 5                    | To rank smartphones according to user-defined priorities and explain the result using actual feature scores.                                             |
| RQ4: Interpretability compared with rating-based recommendation | Objectives 4 and 5; system evaluation | To compare the clarity and usefulness of the feature-based output with a conventional rating-based baseline without introducing an additional objective. |

The fourth research question is evaluated through the system-evaluation phase described in Chapter 3. It does not create an additional research objective because the approved proposal contains five objectives. The comparison examines whether recommendations produced through Objectives 4 and 5 are more understandable than a basic rating-based ranking.

## **1.7 Significance of the Research**

The study contributes to consumer decision support, recommendation-system design, applied ABSA, and explainable artificial intelligence by linking feature-level review evidence directly to transparent smartphone ranking.

### **1.7.1 Significance for Smartphone Buyers**

Buyers can compare feature profiles instead of reading large numbers of reviews manually. Because the ranking reflects explicit priorities, photography-focused, battery-focused, performance-focused, and price-focused users can receive different results from the same product dataset. Visible feature contributions and trade-offs may also improve decision confidence (Dietz et al., 2022).

### **1.7.2 Significance for Recommendation-System Research**

The study offers an auditable alternative to methods based mainly on ratings, latent profiles, or interaction history. Review-aware research shows the value of aspect sentiment, while the proposed design makes the relationship between user preferences and product ranking directly visible through feature weights (Liu et al., 2021; Haghighi & Moradian Zadeh, 2025; Danish et al., 2026). It reduces new-user cold-start dependence, although products without sufficient reviews remain outside the system’s reliable scope.

### **1.7.3 Significance for ABSA and Natural Language Processing**

The research demonstrates how ABSA outputs can be used beyond classification. Feature detection and sentiment labels are converted into product scores, recommendations, and explanations using real smartphone review language. This creates an applied NLP pipeline in which accuracy and practical decision support are evaluated together.

### **1.7.4 Significance for Explainable Artificial Intelligence**

The explanation module is grounded in the feature scores and user weights that determine the ranking. It therefore provides user-centered reasons-why a phone fits the buyer’s priorities rather than only model-centered information about influential tokens or internal predictions (Mansilla et al., 2025; Chen et al., 2026).

### **1.7.5 Practical and Broader Value**

Aggregated feature sentiment can also help manufacturers and e-commerce analysts identify commonly praised or criticized product attributes. The same pipeline could later be adapted to other multi-feature products by changing the aspect categories and domain vocabulary.

## **1.8 Scope and Delimitations of the Research**

The study is limited to the design and evaluation of a research prototype for explainable smartphone recommendation. The boundaries are defined to keep the implementation feasible and to ensure that the system can be evaluated using a stable and traceable dataset.

### **1.8.1 Data Source and Collection Boundary**

The main data source is publicly visible Amazon smartphone review text. Reviews will be collected before model development and evaluation and stored as an offline dataset. The system will not scrape Amazon in real time whenever a user requests a recommendation. This boundary supports reproducibility and prevents changes in live pages from affecting experimental results.

Review text is treated as secondary user-generated data. Private profiles, login-protected information, payment-restricted content, and personal purchasing histories are outside the scope. Reviewer names and identifiers are not required for the analysis.

### **1.8.2 Product and Language Boundary**

The product category is limited to smartphones. Accessories, tablets, laptops, smartwatches, and other electronics are excluded. Restricting the study to one product category allows the aspect vocabulary, feature scoring, and evaluation procedures to remain consistent.

Only English-language reviews will be used in the main implementation. Multilingual ABSA research demonstrates that cross-language review analysis is possible, but it also introduces challenges related to language imbalance, weak supervision, translation, and domain transfer (Satria et al., 2025; Ratniasih et al., 2025). A multilingual extension is therefore reserved for future work.

### 1.8.3 Feature Boundary

The ABSA process is limited to six predefined features: camera, battery, display, performance, design, and price. These features form the feature-scoring and weighting structure approved in the research proposal. Other aspects such as speakers, biometric sensors, software updates, charging speed, network reception, and after-sales service are not included in the primary version.

### 1.8.4 Recommendation Boundary

The recommendation engine will use review-derived feature scores and explicit user-defined feature weights. It will not use collaborative filtering, user similarity, long-term behaviour tracking, or purchase-history profiling. Overall star ratings may be retained as supplementary information and used to construct a basic baseline for comparison, but they will not determine the proposed feature-based ranking.

The system addresses user cold-start by allowing a new user to state feature priorities. It does not claim to solve the new-item cold-start problem for products that have no reviews, because feature performance scores cannot be calculated without sufficient review evidence.

### 1.8.5 Evaluation Boundary

Evaluation will focus on the reliability of feature-level sentiment classification, the relevance of the generated rankings, and the clarity and usefulness of the explanations. Classification will be assessed using a manually labelled test sample and standard metrics such as accuracy, precision, recall, and F1-score. The proposed ranking and explanation output will be compared with a basic overall-rating baseline. The study will not predict future sales, commercial demand, or smartphone market prices.

**Table 1.2: Scope and Delimitations of the Study**

| Area        | Included in the Study                                                             | Not Included in the Study                                                                 |
|-------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Data source | Publicly visible Amazon smartphone review text collected into an offline dataset. | Private user data, login-protected content, and real-time scraping during recommendation. |
| Language    | English-language reviews.                                                         | Multilingual and cross-language model evaluation.                                         |

| Area                 | Included in the Study                                                                       | Not Included in the Study                                                                |
|----------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Product category     | Smartphones.                                                                                | Accessories, tablets, laptops, smartwatches, and other electronics.                      |
| Features             | Camera, battery, display, performance, design, and price.                                   | Additional device-specific aspects outside the approved framework.                       |
| Recommendation logic | ABSA-derived feature scores and user-defined feature weights.                               | Collaborative filtering, user-similarity modelling, and long-term behavioral profiling.  |
| Evaluation           | ABSA metrics, ranking relevance, explanation clarity, and rating-based baseline comparison. | Sales forecasting, market-demand prediction, and production-level commercial deployment. |

## 1.9 Expected Outcomes

The principal expected outcome is a functioning research prototype that demonstrates how unstructured Amazon smartphone reviews can be transformed into feature-level evidence and used to produce explainable recommendations.

1. A clean, organized dataset of Amazon smartphone reviews that is prepared for natural language processing.
2. A functional ABSA pipeline that correctly categorizes sentiment at the feature level from actual Amazon reviews.
3. A feature performance profile that displays all six of the targeted features' percentage positive scores for every smartphone.
4. A useful recommendation engine that generates a ranked, weighted list of recommendations based on user feature priorities.
5. A user-friendly, plain-language explanation produced by AI for every suggestion.
6. When compared to a rating-based baseline, evaluation results show sentiment classification accuracy and recommendation relevance.
7. A comprehensive research report that details the project's design, execution, results, and conclusions.
8. A tidy, organized dataset of Amazon smartphone reviews is prepared for NLP analysis.

## **1.10 Structure of the Dissertation**

The dissertation contains six chapters. Chapter 1 introduces the study. Chapter 2 reviews the selected literature, reports the systematic-review process, identifies the research gap, and presents the conceptual framework. Chapter 3 explains the pragmatic, predominantly quantitative design-and-development methodology, including data collection, ABSA, scoring, ranking, explanation, evaluation, quality assurance, and ethics.

## **1.11 Chapter Summary**

This chapter established smartphone selection as a multi-feature decision for which overall ratings and interaction-based recommendations may provide insufficient detail. It introduced Amazon reviews as experiential evidence and ABSA as the method used to separate opinions about camera, battery, display, performance, design, and price.

The chapter also stated the research problem, aim, five approved objectives, four research questions, significance, scope, and expected outcomes. Chapter 2 now evaluates the selected literature and develops the conceptual framework for the proposed system.

# Chapter 2: Literature Review

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## 2.1 Chapter Introduction

This chapter reviews research supporting an explainable feature-based smartphone recommendation system using ABSA. The central problem is that users receive many ratings and reviews but limited evidence showing whether a phone fits the features they value. Conventional systems can perform well when interaction data are available, yet they may retain cold-start limitations, hide feature trade-offs, and provide weak explanations (Musto et al., 2017; Dang et al., 2021; Liu et al., 2021; Danish et al., 2026).

The review therefore focuses on recommendation systems, online reviews, ABSA methods, sentiment-aware recommendation, explainability, and smartphone or mobile-review analysis. It also reports the systematic-review process for the finalized 32-paper corpus, compares the major methods, identifies the research gap, and presents the conceptual framework.

## 2.2 Product Recommendation Systems

### 2.2.1 Traditional Recommendation Approaches

Collaborative filtering recommends items from user-item interaction patterns, content-based filtering matches item characteristics to user profiles, and hybrid systems combine several signals. These approaches reduce information overload, but they may not reveal which smartphone features explain a ranking and can be limited by sparse or unavailable interaction histories (Musto et al., 2017; Sami et al., 2024; Danish et al., 2026).

### 2.2.2 Limitations of Rating- and Interaction-Based Recommendation

Cold-start occurs when new users or products provide insufficient interaction data. Content and graph-based models can reduce this limitation by adding item, review, or network

information, although they may still require interaction structures (Volkovs et al., 2017; Lin, 2022).

A further problem is the loss of feature-level detail. Overall ratings may hide repeated complaints about battery, heating, or camera quality, and ranked outputs often provide little explanation. For high-involvement purchases, users need visible reasons and trade-offs rather than only a predicted score (Dietz et al., 2022; Chen et al., 2026; Gajula, 2025).

### **2.2.3 Movement toward Review-Aware and Sentiment-Aware Recommendation**

Review-aware systems use textual opinions to enrich user and product representations. Aspect sentiment has improved multi-criteria, deep, multilingual, and hybrid recommendation models (Musto et al., 2017; Dang et al., 2021; Liu et al., 2021; Danish et al., 2026). However, much of this work emphasizes rating prediction or latent profiles, whereas the present study uses explicit feature scores and user-defined weights for transparent smartphone ranking.

## **2.3 Online User Reviews and Smartphone Product Evaluation**

### **2.3.1 Online Reviews as Product Experience Data**

Online reviews describe real or perceived product experience, including battery endurance, camera performance, display quality, heating, responsiveness, design, and value. These signals complement specifications and provide richer evidence than star ratings alone. Review-derived aspect and sentiment information has therefore become an important input to recommendation research (Musto et al., 2017; Dang et al., 2021; Haghighi & Moradian Zadeh, 2025).

### **2.3.2 Importance of Feature-Level Analysis in Smartphone Recommendation**

Smartphones combine strengths and weaknesses across several attributes, so the same product can suit one user and disappoint another. ABSA preserves these differences by assigning sentiment separately to each feature; for example, display-positive and battery-negative can coexist within one sentence (Soni et al., 2024; Singhal et al., 2025; Kumari et al., 2025).

### **2.3.3 Review Credibility and Data Quality**

Review-based scoring is sensitive to duplicate, promotional, spam, and low-information content. Spam-aware research shows that misleading reviews can affect rankings and trust (Jin et al., 2026). The proposed system therefore includes deduplication, language and relevance screening, traceable source records, and minimum evidence requirements.

## **2.4 Aspect-Based Sentiment Analysis**

### **2.4.1 Concept and Core Tasks of ABSA**

ABSA identifies product aspects and the sentiment expressed toward each one. Its tasks range from aspect and opinion extraction to aspect-sentiment pairing, triplet extraction, and aspect-category sentiment classification (Aziz et al., 2024; Maldonado Castillo et al., 2024). In this study, the output is structured feature sentiment that can be aggregated for recommendation.

### **2.4.2 Lexicon-Based and Classical Machine Learning Approaches**

Classical ABSA uses lexicons, part-of-speech information, dependency rules, feature selection, and classifiers such as Naïve Bayes. These methods are efficient and interpretable but depend on feature engineering and may struggle with implicit or context-dependent sentiment (Mubarok et al., 2017). Recent syntactic approaches remain useful for transparent

aspect-opinion mapping, although contextual models generally handle noisy reviews more effectively (Mansilla et al., 2025).

### **2.4.3 Neural Network and Hybrid Deep Learning Approaches**

CNN, recurrent, and BiLSTM models learn local and sequential patterns from review text, while hybrid models combine contextual embeddings with sequence or task-specific components. Studies using CNN and Word2Vec, hybrid e-commerce classifiers, BERT-BiLSTM, multimodal text-emoji features, and multitask learning demonstrate improved representation but also highlight data, tuning, and class-imbalance requirements (Lestari et al., 2024; Rasappan et al., 2024; Setiabudi & Faizah, 2025; Padigapati & Krishna, 2025; Chong et al., 2026).

### **2.4.4 Transformer-Based ABSA Models**

Transformers such as BERT, DeBERTa, and XLM-RoBERTa model words in context, which is valuable when smartphone terms change meaning across features or sentences contain several opinions. Unified BERT-graph models and Instruct-DeBERTa illustrate how contextual representations can support extraction and classification together (Aziz et al., 2024; Jayakody et al., 2024).

Mobile-phone studies also report stronger performance for BERT than several classical and neural baselines (Soni et al., 2024; Singhal et al., 2025). Transformer-based ABSA is therefore the strongest fit for the core classifier, while transparent aspect categories and validation remain necessary.

### **2.4.5 Multi-Domain and Multilingual ABSA**

ABSA performance can decline across products, platforms, and languages. Multilingual studies using XLM-RoBERTa and contextual models show the potential for cross-language analysis but also expose imbalance, supervision, and transfer challenges (Satria et al., 2025; Ratniasih et al., 2025; Di Nuovo et al., 2024). The present system is therefore limited to English Amazon smartphone reviews, with multilingual extension reserved for future work.

## **2.5 Integration of ABSA with Recommendation Systems**

### **2.5.1 Aspect Sentiment as Recommendation Evidence**

Aspect sentiment can function as structured recommendation evidence. Multi-criteria and review-aware studies show that it can represent product strengths and user preferences more meaningfully than one overall rating (Musto et al., 2017; Liu et al., 2021; Haghighi & Moradian Zadeh, 2025). The proposed system makes this evidence visible through product feature scores.

### **2.5.2 Multi-Criteria and Feature-Based Recommendation**

Each smartphone feature can be treated as an independent recommendation criterion. User-defined weights then determine how camera, battery, display, performance, design, and price contribute to the final score. This explicit approach supports personalization and makes feature trade-offs understandable (Dietz et al., 2022; Chabane et al., 2023).

### **2.5.3 Cold-Start Handling through Review-Derived Product Profiles**

Content and review signals can support recommendation when interaction histories are limited (Volkovs et al., 2017; Lin, 2022; Danish et al., 2026). The proposed system addresses new-user cold-start by collecting feature priorities at the start of the session. New products can be included only after sufficient review evidence is available.

### **2.5.4 Related Applications beyond Smartphone Recommendation**

Studies in healthcare, grocery shopping, tourism, and general e-commerce show that opinion mining, product features, and hybrid models can support decision-oriented recommendation across domains (Keikhosrokiani et al., 2024; Chabane et al., 2023; Dietz et al., 2022; Gajula, 2025). Their methods are transferable, although smartphone-specific vocabulary and evaluation remain necessary.

## **2.6 Explainable Recommendation and User Trust**

### **2.6.1 Need for Explainability in Smartphone Selection**

Smartphones are high-involvement purchases, so users need to understand why a device appears in the recommended list. Explanations can improve perceived usefulness and trust when they reveal the features and trade-offs behind a ranking (Dietz et al., 2022; Chen et al., 2026; Gajula, 2025).

### **2.6.2 Explanation Generation from Feature Scores**

Feature-based explanations can identify the highest weighted contributions and cite the supporting scores or mention counts. Aspect-opinion-sentiment extraction and interpretable ABSA demonstrate how structured evidence can be converted into concise user-facing reasons (Maldonado Castillo et al., 2024; Mansilla et al., 2025; Danish et al., 2026; Zhu, 2025).

### **2.6.3 Model Interpretability and User-Centered Explanation**

Model-centered interpretability describes how a prediction was produced, whereas user-centered explanation states why the recommended product suits the buyer. The proposed system prioritizes the latter by connecting visible feature scores and user weights to the final ranking, while model-level methods such as causal reasoning or SHAP remain useful for technical inspection (Chen et al., 2026; Zhu, 2025).

## **2.7 Smartphone and Mobile Review Analysis**

Mobile-phone studies provide the closest empirical support for this research. Web-scraped phone reviews contain usable aspect signals, and BERT-based models have outperformed SVM, Naïve Bayes, CNN, RNN, and other classical methods in relevant studies (Soni et al., 2024; Singhal et al., 2025). Multimodal work also shows that emojis and informal expressions can influence aspect sentiment (Padigapati & Krishna, 2025).

Related mobile-app studies highlight preprocessing noise, class imbalance, multilinguality, and the need for contextual models (Lestari et al., 2024; Pranatawijaya et al., 2024;

Ratniasih et al., 2025). These issues inform the proposed data-cleaning, annotation, model-selection, and evaluation procedures.

## 2.8 Systematic Literature Review Process

### 2.8.1 Review Questions

The systematic literature review was conducted to identify how existing studies support the design of an explainable feature-based smartphone recommendation system using ABSA.

The review was guided by four questions:

- How can sentiment at the feature level be reliably extracted from smartphone and product reviews?
- How can feature-level sentiment scores be converted into useful product performance metrics?
- How can explainable recommendations be generated using feature scores and user preferences?
- Compared with conventional rating-based systems, how can feature-based recommendation improve interpretability?

### 2.8.2 Search Strategy and Information Sources

The PRISMA-style search used IEEE Xplore, Springer Nature Link, ResearchGate, and Google Scholar. Searches conducted from 11 to 22 June 2026 covered ABSA, smartphone and product reviews, sentiment-aware and explainable recommendation, and transformer or machine-learning methods. Final verification was completed on 25 June 2026.

**Table 2.1: Search sources, search dates, and records retrieved**

| <b>Databa<br/>se /<br/>Source</b> | <b>Search<br/>date</b> | <b>Search<br/>fields<br/>used</b> | <b>Main filters applied</b>                            | <b>Records<br/>retrieved</b> |
|-----------------------------------|------------------------|-----------------------------------|--------------------------------------------------------|------------------------------|
| IEEE<br>Xplore                    | 22 June<br>2026        | Title, abstract,<br>keywords      | English; journal/conference<br>papers; ABSA, sentiment | 86                           |

| <b>Database / Source</b> | <b>Search date</b> | <b>Search fields used</b>  | <b>Main filters applied</b>                                                                               | <b>Records retrieved</b> |
|--------------------------|--------------------|----------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------|
|                          |                    |                            | analysis, recommender systems, explainability                                                             |                          |
| Springer Nature Link     | 13 June 2026       | Title, abstract, keywords  | English; computer science, information systems, AI, NLP, recommendation systems                           | 1,213                    |
| ResearchGate             | 15 June 2026       | Title and full-text search | Topic relevance, full-text availability, ABSA/recommendation relevance                                    | 53                       |
| Google Scholar           | 18 June 2026       | Title and keyword search   | 2017-2026; relevance to ABSA, product reviews, sentiment-aware recommendation, explainable recommendation | 25                       |
| Total                    |                    |                            |                                                                                                           | 1,377                    |

### 2.8.3 Search Strings

A composite Boolean string combined ABSA and sentiment terms with smartphone or product reviews, recommendation-system terms, and transformer or machine-learning methods. Shorter combinations were used where database interfaces imposed search limitations.

### 2.8.4 Inclusion and Exclusion Criteria

The criteria retained English academic studies that supported at least one component of the proposed framework. The main publication period was 2021–2026, with selected foundational 2017 studies retained for ABSA, multi-criteria recommendation, and cold-start evidence.

**Table 2.2: Inclusion and exclusion criteria**

| Type      | Code | Criteria                                                                                                                                                                                                                                  |
|-----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inclusion | IC1  | Studies published mainly between 2021 and 2026 were included; foundational earlier works were retained where necessary.                                                                                                                   |
| Inclusion | IC2  | Studies written in English were included.                                                                                                                                                                                                 |
| Inclusion | IC3  | Studies related to ABSA, sentiment analysis, opinion mining, product reviews, smartphone reviews, mobile app reviews, or customer reviews were included.                                                                                  |
| Inclusion | IC4  | Studies related to recommender systems, review-aware recommendation, sentiment-aware recommendation, feature-based recommendation, multi-criteria recommendation, cold-start recommendation, or explainable recommendation were included. |
| Inclusion | IC5  | Studies using machine learning, deep learning, BERT, DeBERTa, XLM-RoBERTa, CNN, LSTM, graph-based models, hybrid models, or transformer-based sentiment models were included.                                                             |
| Inclusion | IC6  | Studies with accessible full text or enough bibliographic information for DOI/source verification were included.                                                                                                                          |
| Inclusion | IC7  | Studies that directly or indirectly supported the proposed ABSA-driven smartphone recommendation framework were included.                                                                                                                 |
| Exclusion | EC1  | Studies unrelated to ABSA, sentiment analysis, user reviews, product reviews, smartphone/mobile reviews, or recommender systems were excluded.                                                                                            |
| Exclusion | EC2  | Duplicate records and repeated studies with the same DOI, title, or author-topic combination were excluded.                                                                                                                               |
| Exclusion | EC3  | Studies without accessible full text were excluded if their methods and findings could not be verified.                                                                                                                                   |
| Exclusion | EC4  | Non-academic sources such as blogs, marketing websites, general web articles, and unsupported online content were excluded.                                                                                                               |
| Exclusion | EC5  | Studies focusing only on general rating prediction without review text, feature-level sentiment, or explainability were excluded.                                                                                                         |

| Type      | Code | Criteria                                                                        |
|-----------|------|---------------------------------------------------------------------------------|
| Exclusion | EC6  | Studies outside the scope after full-text eligibility assessment were excluded. |

## 2.8.5 Article Selection Process

The search identified 1,377 records. After duplicate removal, automated and other exclusions, screening, retrieval, and eligibility assessment, 32 studies remained for synthesis.

PRISMA 2020 flow diagram for new systematic reviews which included searches of databases and registers only

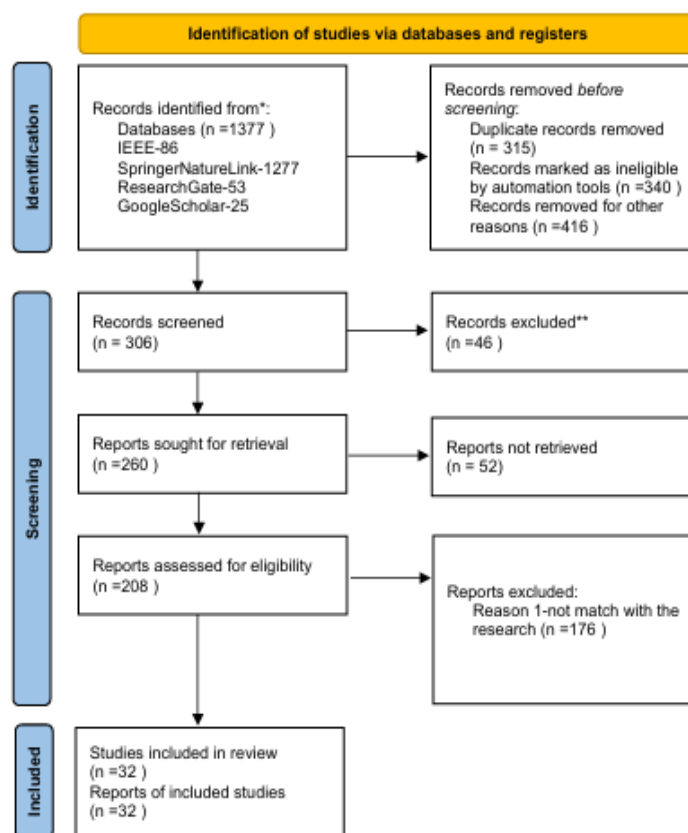


Figure 2.1: PRISMA-style article selection process

## 2.8.6 Data Extraction and Thematic Coding

Each paper was coded by focus, domain or dataset, method, main findings, limitations, and relevance to review collection, ABSA, scoring, recommendation, explainability, cold-start, multilinguality, or credibility.

**Table 2.3: Thematic classification of the selected 32-paper corpus**

| <b>Theme</b> | <b>Focus</b>                                                                 | <b>Number of studies</b> |
|--------------|------------------------------------------------------------------------------|--------------------------|
| Theme 1      | ABSA methods and transformer-based sentiment classification                  | 15                       |
| Theme 2      | Review-aware and sentiment-aware recommendation                              | 8                        |
| Theme 3      | Feature-based, multi-criteria, conversational, and cold-start recommendation | 5                        |
| Theme 4      | Smartphone/mobile review evidence, multilinguality, and domain transfer      | 4                        |
| Total        |                                                                              | 32                       |

### **2.8.7 Classification of the Selected Papers**

Table 2.4 classifies the 32 studies and shows the specific contribution of each paper to the proposed framework.

**Table 2.4: Classification of the selected 32 studies**

| No | Study                  | Main focus                                    | Method / dataset                                                                          | Relevance to proposed system                                                                     |
|----|------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| 1  | Musto et al. (2017)    | Multi-criteria recommender systems using ABSA | Aspect sentiment from user reviews integrated into multi-criteria collaborative filtering | Justifies converting aspect sentiment into recommendation criteria.                              |
| 2  | Mubarak et al. (2017)  | Classical ABSA                                | POS tagging, Chi-Square feature selection, and Naïve Bayes on product reviews             | Provides a baseline for early supervised ABSA methods.                                           |
| 3  | Volkovs et al. (2017)  | Cold-start recommendation                     | Content-based neighbor models for recommender system challenge data                       | Supports content-based signals when interaction history is limited.                              |
| 4  | Liu et al. (2021)      | Multilingual review-aware recommendation      | MABSA, multilingual embeddings, and dual interactive attention                            | Shows that aspect sentiment can improve recommendation and interpretability.                     |
| 5  | Dang et al. (2021)     | Sentiment-aware recommendation                | BERT feature extraction, deep sentiment models, and collaborative filtering               | Supports integration of sentiment analysis with recommendation algorithms.                       |
| 6  | Lin (2022)             | Cold-start recommendation                     | User-interaction parallel networks using user, item, social, and review features          | Supports the use of review information for cold-start contexts.                                  |
| 7  | Dietz et al. (2022)    | Feature trade-off recommendation              | Trade-off visualization in content-based recommendation                                   | Supports showing feature trade-offs to improve user understanding.                               |
| 8  | Chabane et al. (2023)  | Feature-based product recommendation          | Clustering and machine learning for grocery recommendation                                | Shows wider applicability of feature-based product recommendation.                               |
| 9  | Aziz et al. (2024)     | Unified transformer ABSA                      | BERT, graph convolution, biaffine attention, and multiple ABSA subtasks                   | Supports transformer and graph-enhanced ABSA for aspect extraction and sentiment classification. |
| 10 | Rasappan et al. (2024) | E-commerce sentiment classification           | Hybrid deep learning with term weighting and feature selection                            | Supports preprocessing and hybrid deep learning for product reviews.                             |

| No | Study                              | Main focus                               | Method / dataset                                                             | Relevance to proposed system                                                        |
|----|------------------------------------|------------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 11 | Di Nuovo et al. (2024)             | Multilingual sentiment analysis          | XLM-R-based sentiment model for news data                                    | Supports multilingual transformer adaptation as a future direction.                 |
| 12 | Keikhosrokiani et al. (2024)       | Opinion-mining recommendation            | Opinion mining and machine learning for drug recommendation                  | Shows that review sentiment can support decision-oriented recommendation.           |
| 13 | Lestari et al. (2024)              | Mobile app review ABSA                   | CNN and Word2Vec on Google Play reviews                                      | Highlights mobile-review preprocessing and class imbalance issues.                  |
| 14 | Maldonado Castillo et al. (2024)   | Aspect-opinion-sentiment extraction      | Triplet extraction and review synthesis for restaurant reviews               | Supports generating structured explanations from aspect-opinion-sentiment evidence. |
| 15 | Pranatawijaya et al. (2024)        | Mobile app review sentiment              | Topic modeling, aspect generation, SMOTE, and BERT classification            | Supports aspect discovery and class balancing in noisy review data.                 |
| 16 | Sami et al. (2024)                 | Hybrid recommendation                    | Deep learning hybrid model using CF, RNN, NCF, and content analysis          | Supports combining multiple recommendation signals.                                 |
| 17 | Soni et al. (2024)                 | Smartphone/mobile phone ABSA             | Web scraping and ABSA of Flipkart mobile phone reviews; BERT comparison      | Directly supports web-scraped mobile phone review ABSA.                             |
| 18 | Gajula (2025)                      | Sentiment-aware recommendation review    | NLP perspective on sentiment-aware e-commerce recommendation                 | Supports the theoretical value of sentiment in recommendation and explanation.      |
| 19 | Haghighi and Moradian Zadeh (2025) | Aspect-aware review-based recommendation | DeBERTa aspect-sentiment profiles and interaction modeling                   | Supports using aspect-sentiment profiles in recommendation.                         |
| 20 | Jayakody et al. (2024)             | Instruction-based transformer ABSA       | InstructABSA with DeBERTa for aspect extraction and sentiment classification | Supports pipelined aspect extraction and transformer sentiment classification.      |
| 21 | Kumari et al. (2025)               | Product review ABSA                      | Aspect-based sentiment analysis for product reviews                          | Supports feature-level product evaluation from customer reviews.                    |
| 22 | Mansilla et al. (2025)             | Interpretable ABSA                       | Syntactic patterns, dependency parsing, and VADER sentiment scoring          | Supports interpretable aspect-sentiment mapping.                                    |

| No | Study                         | Main focus                                        | Method / dataset                                                      | Relevance to proposed system                                                       |
|----|-------------------------------|---------------------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 23 | Padigapati and Krishna (2025) | Smartphone multimodal ABSA                        | Text and emoji-based ABSA using Word2Vec, Emoji2Vec, BERT, and BiLSTM | Supports handling informal smartphone review signals.                              |
| 24 | Ratniasih et al. (2025)       | Mobile app sentiment review                       | Review of multilingual and contextual sentiment techniques            | Supports transformer shift and multilingual limitations in mobile review analysis. |
| 25 | Satria et al. (2025)          | Multilingual customer product ABSA                | ABAE, FastText/MUSE, and XLM-RoBERTa on multilingual product reviews  | Supports multilingual ABSA as a future extension.                                  |
| 26 | Setiabudi and Faizah (2025)   | Hybrid neural ABSA                                | BERT embeddings and BiLSTM on FABSA dataset                           | Supports contextual-sequential ABSA modeling.                                      |
| 27 | Singhal et al. (2025)         | Mobile phone technical review sentiment           | BERT-based sentiment classification compared with classical models    | Directly supports mobile phone technical sentiment modeling.                       |
| 28 | Zhu (2025)                    | Transformer ABSA explainability                   | DeBERTa-v3, LoRA fine-tuning, and SHAP interpretability               | Supports interpretable transformer-based sentiment modeling.                       |
| 29 | Chen et al. (2026)            | Explainable recommendation                        | Causal neuro-symbolic reasoning for multi-behavior recommendation     | Supports explainable recommendation logic.                                         |
| 30 | Chong et al. (2026)           | Multitask ABSA                                    | BERT embeddings, BiLSTM, self-attention, and subjectivity detection   | Supports separating subjective review opinions from objective statements.          |
| 31 | Danish et al. (2026)          | Sentiment-aware explainable hybrid recommendation | SBERT, ABSA, collaborative filtering, and SHAP explanations           | Supports end-to-end sentiment-aware and explainable recommendation.                |
| 32 | Jin et al. (2026)             | Spam-aware recommendation                         | Spam review detection and credibility-aware recommendation            | Supports review credibility filtering as a quality-control concern.                |

### 2.8.8 Data Synthesis

The synthesis links four evidence streams. ABSA research supports aspect and sentiment extraction; recommendation studies support the use of aspect-level evidence in ranking; explainability research supports visible feature-based reasons; and cold-start research supports content and explicit preference signals when user history is limited. Transformer models provide the strongest contextual fit, but the literature also confirms that classification accuracy must be connected to interpretable product scoring and user-facing decision support.

## 2.9 Critical Comparison of Reviewed Methods

The reviewed methods differ in contextual accuracy, interpretability, computational cost, data requirements, and suitability for recommendation. Classical approaches are efficient and transparent but rely on feature engineering; neural and transformer models capture richer context but require tuning and validation. Review-aware recommendation uses evidence beyond ratings, yet many systems retain latent profiles or model-centred explanations. Table 2.5 summarizes the fit of each method group for this study.

**Table 2.5: Critical comparison of reviewed method groups**

| <b>Method group</b>         | <b>Strengths</b>                                                 | <b>Limitations</b>                                                                 | <b>Fit for current research</b>                                                   |
|-----------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Lexicon and rule-based ABSA | Interpretable, simple, and able to link aspects to opinion terms | Weak with sarcasm, implicit sentiment, domain-specific language, and noisy reviews | Useful for aspect dictionaries and explanation support, but not sufficient alone. |
| Classical machine learning  | Efficient and workable with smaller labeled datasets             | Requires feature engineering and has limited contextual understanding              | Useful as baseline but weaker for mixed smartphone review sentences.              |

| Method group                                    | Strengths                                                                              | Limitations                                                                           | Fit for current research                                                       |
|-------------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| Neural and hybrid deep learning                 | Learns semantic and sequential patterns; can handle richer representations             | Requires training data, tuning, and imbalance handling                                | Useful where transformer embeddings are combined with sequence models.         |
| Transformer-based ABSA                          | Strong contextual understanding and effective for technical mixed-opinion reviews      | Higher computational cost and requires domain adaptation and validation               | Most suitable for the core ABSA component of the proposed system.              |
| Review-aware and sentiment-aware recommendation | Uses textual evidence beyond ratings and improves personalization and interpretability | Often focuses on rating prediction or latent profiles rather than transparent scoring | Suitable foundation, but needs explicit user weighting and explanation.        |
| Explainable recommendation                      | Improves user trust and decision confidence                                            | Some explanations are model-centered or generic                                       | Directly relevant because the proposed system uses feature-score explanations. |
| Spam-aware and credibility-aware recommendation | Improves reliability of review-based recommendations                                   | Requires labeled spam or credibility data and extra processing                        | Useful as a data-quality extension for Amazon review analysis.                 |

### 2.9.1 Comparative Performance of ABSA Models

The evidence shows a progression from Naïve Bayes and engineered features to neural, hybrid, and transformer architectures. Recent BERT- and DeBERTa-based studies provide

the strongest contextual performance for multi-aspect and technical review language (Mubarak et al., 2017; Aziz et al., 2024; Jayakody et al., 2024; Soni et al., 2024; Singhal et al., 2025; Chong et al., 2026).

### **2.9.2 Limitations of Existing Approaches**

Three limitations remain: ABSA studies often stop at classification; recommender studies frequently prioritize rating prediction or latent interactions; and explanation methods may not directly connect review evidence, user priorities, and product rank. These limitations define the integration requirement of the proposed system.

## **2.10 Research Gap and Positioning of the Study**

The selected literature establishes ABSA, transformer sentiment classification, review-aware recommendation, and explainability, but their full smartphone-specific integration remains limited. Mobile-phone studies extract useful feature sentiment, yet they do not generally build a transparent user-weighted smartphone ranking engine (Soni et al., 2024; Padigapati & Krishna, 2025; Singhal et al., 2025).

A second gap concerns preference representation. Aspect-aware recommenders show that feature sentiment can improve prediction, but many rely on collaborative filtering, learned profiles, or neural interactions (Musto et al., 2017; Liu et al., 2021; Haghighi & Moradian Zadeh, 2025). The proposed system instead uses explicit feature weights and an auditable scoring formula.

A third gap is user-interface explanation. Existing work provides syntactic, attribution, trade-off, or neuro-symbolic explanations, but fewer systems state in simple language why a particular smartphone matches the user’s priorities using actual feature scores (Dietz et al., 2022; Mansilla et al., 2025; Zhu, 2025; Chen et al., 2026).

The study is therefore positioned as an applied design-and-development project integrating Amazon review collection, preprocessing, ABSA, feature scoring, user weighting, ranking, and evidence-based explanation in one smartphone prototype.

**Table 2.6: Research gaps and positioning of the proposed study**

| <b>Identified gap</b>                | <b>Evidence from selected literature</b>                                                                                                                                                                                 | <b>How the proposed study addresses it</b>                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Gap 1: Smartphone specificity        | Most ABSA and recommendation studies use restaurants, laptops, apps, books, movies, groceries, healthcare, tourism, or general e-commerce datasets. Smartphone-specific ABSA-recommendation integration remains limited. | Uses Amazon smartphone reviews and predefined smartphone features: camera, battery, display, performance, design, and price.          |
| Gap 2: User-weighted feature scoring | Many systems use latent profiles, ratings, collaborative filtering, or neural interaction models rather than explicit user feature priorities.                                                                           | Uses a transparent weighted scoring formula where user-selected feature weights are multiplied by review-derived feature scores.      |
| Gap 3: Explanation generation        | Existing explanations are often model-centered, generic, or not linked to actual feature sentiment percentages.                                                                                                          | Generates plain-language explanations grounded in feature sentiment scores and user-selected priorities.                              |
| Gap 4: End-to-end integration        | ABSA, recommendation, cold-start handling, data scraping, scoring, and explainability are often studied separately.                                                                                                      | Integrates Amazon scraping, preprocessing, ABSA, feature scoring, user-weighted ranking, and explanation generation in one prototype. |

## 2.11 Proposed Conceptual Framework

The conceptual framework connects the literature to the system design. Amazon review text is processed into aspect-level sentiment evidence and feature scores; user-defined feature weights express personal priorities; weighted contributions generate the recommendation score and rank; and the interface provides feature-based explanations. Evaluation examines ABSA performance, ranking relevance, and explanation usefulness.

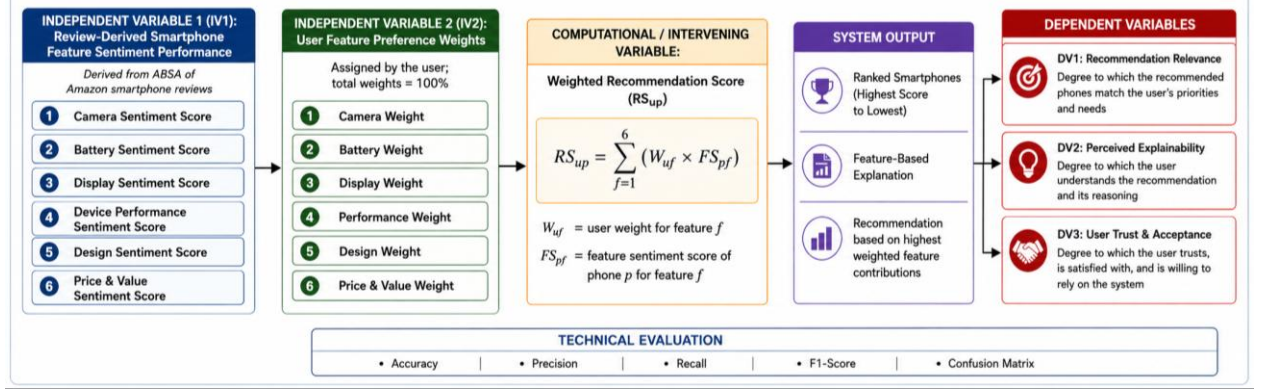


Figure 2.2: Proposed conceptual framework for the ABSA-driven smartphone recommendation system

Source: Developed by the researcher.

## 2.12 Chapter Summary

The review confirms that ABSA and transformer models can recover feature-level opinion from noisy product text, while review-aware, multi-criteria, cold-start, and explainability studies support explicit feature evidence and visible trade-offs in recommendation. The remaining gap is a cohesive smartphone-specific system that converts Amazon reviews into transparent feature scores, user-weighted rankings, and plain-language explanations. The conceptual framework addresses this gap and guides the methodology in Chapter 3.

# Chapter 3: Methodology

---

## 3.1 Chapter Introduction

This chapter explains the methodology used to construct and evaluate the proposed smartphone recommendation artefact. The study collects publicly visible English-language reviews from selected Amazon smartphone listings. The six features-camera, battery, display, performance, design, and price are identified from the review text itself rather than taken from manufacturer specifications. The extracted feature opinions are classified as positive, negative, or neutral, aggregated into product-level feature scores, combined with user-defined feature weights, and converted into ranked recommendations with traceable explanations.

The methodology covers Amazon dataset construction, preprocessing, aspect and sentiment annotation, hybrid aspect identification, transformer-based sentiment classification, feature-score calculation, weighted ranking, explanation generation, and offline technical evaluation. No questionnaire, interview, or participant-based user study is included. The design is informed only by the selected SLR evidence on smartphone review analysis, transformer ABSA, sentiment-aware recommendation, and explainability (Soni et al., 2024; Aziz et al., 2024; Musto et al., 2017; Haghighi & Moradian Zadeh, 2025).

## 3.2 Research Philosophy

The study adopts pragmatism because methods are selected according to their usefulness in solving the smartphone decision-support problem. Amazon review text is treated as practical evidence, processed through NLP and aspect-level sentiment classification, transformed into numerical feature scores, and evaluated through measurable technical criteria.

Pragmatism also supports iterative artefact development followed by controlled evaluation on frozen, unseen review data. Development data are used to refine aspect rules, model settings, and explanation templates, while held-out review data are used to assess the final system.

### **3.3 Research Approach and Type of Study**

The research is predominantly quantitative and computational. Review text is converted into aspect categories, sentiment labels, feature scores, user weights, recommendation scores, ranks, and performance metrics. The main findings therefore depend on measurable system outputs rather than manual thematic analysis.

It is an applied study because it produces a working prototype for smartphone selection. No human participants are recruited. Evaluation is conducted using manually annotated review data, baseline models, predefined feature-weight profiles, held-out review evidence, and automated consistency checks. Formal causal hypotheses are unnecessary because the purpose is artefact construction and performance evaluation rather than modelling latent social-science relationships.

### **3.4 Research Strategy**

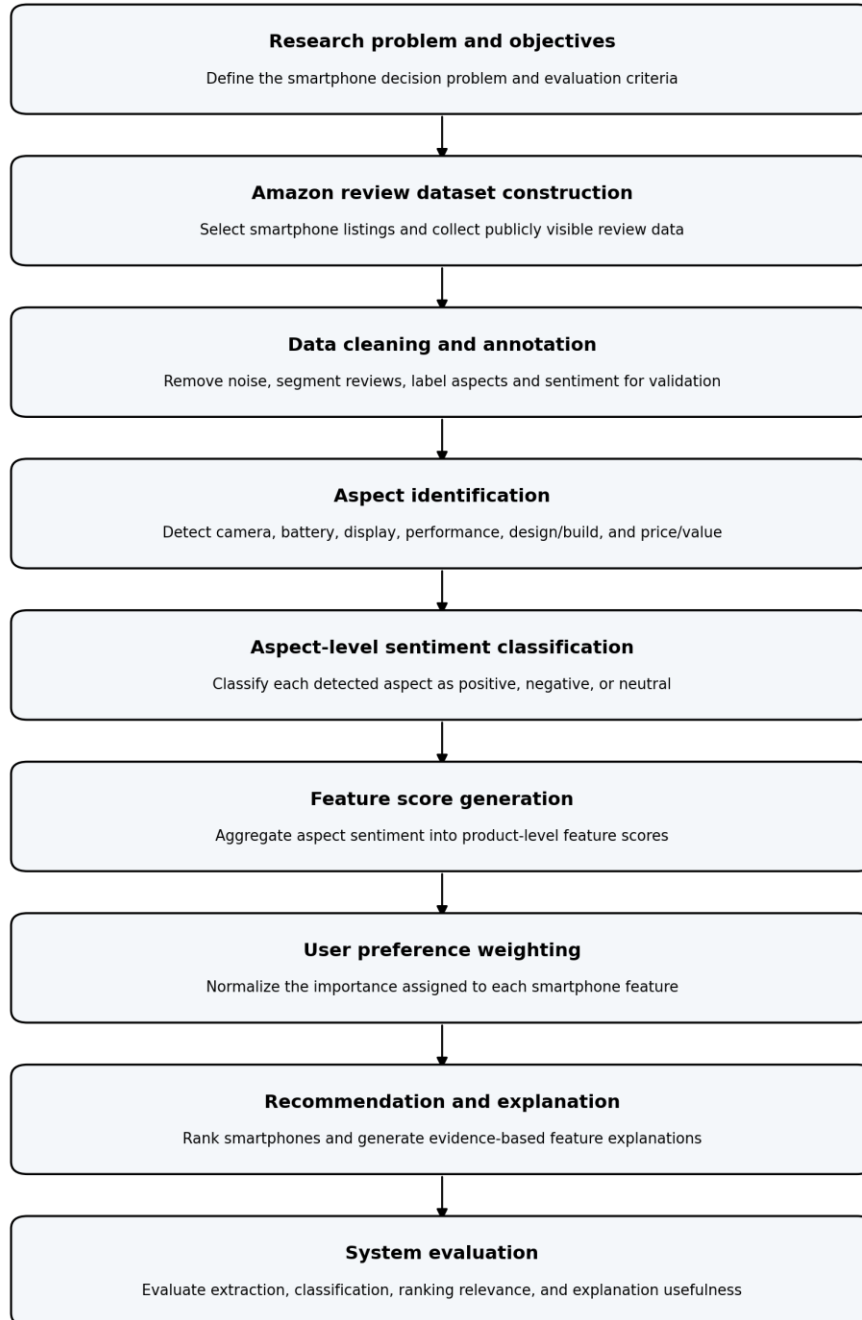
An applied design-and-development strategy is combined with quantitative experimental evaluation. The artefact integrates data collection, ABSA, feature scoring, preference weighting, recommendation, and explanation rather than treating sentiment classification as the final output.

Development is iterative, but tuning and final testing are separated. Training and validation data guide preprocessing rules, the aspect lexicon, model parameters, and explanation templates; the held-out test set is used only after the final configuration is selected.

### **3.5 Research Design and Time Horizon**

The design is sequential and modular: each stage produces a defined output for the next stage, from Amazon review collection to offline system evaluation. Figure 3.1 presents this workflow.

The time horizon is cross-sectional. Reviews are collected during a documented period and frozen as a versioned offline dataset, ensuring that all experiments use the same evidence and can be reproduced.



*Figure 3.1: Research methodology workflow*

*Source: Developed by the researcher.*

**Table 3.1: Overall methodological profile**

| <b>Methodological component</b> | <b>Adopted choice</b>                               | <b>Reason for the choice</b>                                                                           |
|---------------------------------|-----------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Research philosophy             | Pragmatism                                          | Focuses on the practical development and evaluation of a useful decision-support artefact.             |
| Research approach               | Predominantly quantitative and computational        | Transforms review text into measurable labels, scores, rankings, and evaluation metrics.               |
| Research type                   | Applied research                                    | Addresses a practical smartphone-selection problem through a working prototype.                        |
| Research strategy               | Design development and experimental evaluation with | Builds an integrated system and evaluates each component using controlled criteria.                    |
| Time horizon                    | Cross-sectional                                     | Uses a fixed snapshot of Amazon reviews collected during a defined period.                             |
| Main data source                | Secondary user-generated review data                | Amazon reviews were originally created for product feedback, not specifically for this research.       |
| Human participants              | Not included                                        | Evaluation uses annotated Amazon reviews, held-out review data, baseline models, and automated checks. |

**Table 3.2: Alignment between research objectives and methodological procedures**

| <b>Research objective</b>                                  | <b>Methodological procedure</b>                                                                              | <b>Expected output</b>                     |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| RO1: Construct a reliable Amazon smartphone review dataset | Purposive product selection, compliant scraping, eligibility screening, deduplication, and versioned storage | Clean, traceable smartphone review dataset |

| Research objective                                              | Methodological procedure                                                                                     | Expected output                                                                                   |
|-----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| RO2: Identify smartphone features and classify aspect sentiment | Hybrid aspect identification, manual annotation, and BERT-based three-class sentiment classification         | Aspect-sentiment records for camera, battery, display, performance, design/build, and price/value |
| RO3: Generate product-level feature performance scores          | Calculate the percentage of positive review mentions for each smartphone feature                             | Comparable feature scores from 0 to 100                                                           |
| RO4: Rank smartphones according to user priorities              | Normalize feature weights and calculate a weighted recommendation score for each eligible product            | Personalized ranked list of smartphones                                                           |
| RO5: Generate transparent recommendation explanations           | Use feature scores, contribution values, evidence counts, and trade-offs in rule-based explanation templates | Traceable natural-language explanation for each recommendation                                    |

## 3.6 Population, Units of Analysis, and Sample

### 3.6.1 Data Population

The target data population consists of publicly accessible English-language reviews associated with selected smartphone listings on the chosen Amazon marketplace during the defined data-collection period. Since Amazon review content changes over time, the study will use a fixed and documented snapshot of the available review data rather than attempting to include every review ever published.

The reviews are treated as secondary user-generated data because they were originally posted by customers for product feedback and were not created specifically for this research. Reviewer names, profile details, and other personally identifying information are not required for the analysis and will not be collected or retained.

### 3.6.2 Units of Analysis

The primary unit of analysis is the aspect-level clause extracted from each smartphone review. Reviews will be divided into sentences and, where necessary, into separate clauses to ensure that different feature opinions are analysed independently.

For example, the statement, “The camera is excellent, but the battery drains quickly,” will be represented as two separate analytical records:

Camera - positive sentiment

Battery - negative sentiment

The smartphone product is the aggregation unit because all aspect-level sentiment results associated with a particular smartphone will be combined to calculate feature performance scores for camera, battery, display, performance, design, and price.

### 3.6.3 Sample

The computational sample consists of publicly accessible English-language Amazon reviews collected from purposively selected smartphone listings. Approximately 15 smartphone models will represent different brands and price categories. A minimum of 200 accessible textual reviews per model will be targeted, producing an expected raw dataset of at least 3,000 reviews.

After language screening, duplicate removal, and removal of unusable records, approximately 2,400-2,700 reviews are expected to remain. This target provides repeated evidence for camera, battery, display, performance, design, and price and supports the minimum requirement of 10 valid mentions for a product-feature score. It is an operational benchmark rather than a statistically representative sample because review availability differs across Amazon listings. All accessible reviews meeting the criteria will be retained. From the cleaned corpus, at least 2,400 aspect-level clauses will be manually annotated, aiming for approximately 400 clauses per feature and coverage of positive, negative, and neutral sentiment. The labelled data will use the planned 70:15:15 training, validation, and test split. Actual counts of models, raw reviews, retained reviews, and annotated clauses will be reported after collection. No human participants are included.

### 3.7 Sampling Strategy and Eligibility Criteria

Smartphone listings are selected purposively to provide sufficient evidence across the six features and variation in brand and price category. Duplicate configurations are avoided unless they contain distinct review pools.

Reviews are screened by predefined criteria. Public English text linked to an eligible phone is retained, while irrelevant, empty, inaccessible, duplicate, and near-duplicate records are removed. Available credibility indicators are stored for quality checks because misleading reviews can affect recommendation outcomes (Jin et al., 2026).

**Table 3.3: Inclusion and exclusion criteria for the Amazon review dataset**

| Code | Inclusion criteria                                                                           | Code | Exclusion criteria                                                                                |
|------|----------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------|
| IC1  | Publicly visible reviews linked to selected smartphone listings.                             | EC1  | Reviews for accessories, tablets, smartwatches, laptops, or unrelated products.                   |
| IC2  | Reviews written in English.                                                                  | EC2  | Non-English reviews in the main implementation.                                                   |
| IC3  | Reviews containing meaningful textual content.                                               | EC3  | Empty records, rating-only records, and extremely short comments without usable opinion evidence. |
| IC4  | Reviews collected within the documented data-collection procedure.                           | EC4  | Login-protected, restricted, or inaccessible content.                                             |
| IC5  | Reviews containing at least one relevant smartphone aspect or usable contextual evidence.    | EC5  | Reviews with no detectable relationship to the six selected aspects.                              |
| IC6  | Unique review records after exact and near-duplicate screening.                              | EC6  | Exact duplicates, copied reviews, and near-duplicate content likely to inflate sentiment counts.  |
| IC7  | Reviews whose product identity can be verified through an ASIN or stable product identifier. | EC7  | Records whose product cannot be reliably linked to a smartphone model.                            |

## 3.8 Data Collection Procedure

### 3.8.1 Source and Collection Mode

Reviews are collected from publicly visible Amazon smartphone review pages before model development. A Python collector uses Playwright or Selenium for dynamically rendered content and BeautifulSoup where static extraction is sufficient. Only review text and supporting listing metadata are collected; camera, battery, display, performance, design, and price scores are derived later from opinions expressed in the reviews, not copied from product specifications. Collection follows applicable platform rules, controlled request intervals, caching, and error logging (Soni et al., 2024).

### 3.8.2 Collection Steps

- Prepare a product register containing the selected smartphone model, brand, configuration, ASIN, URL, price category, and collection status.
- Open each eligible review page and retrieve the review records available through the permitted interface.
- Extract only the fields needed for product grouping, review analysis, reproducibility, and baseline comparison.
- Store the unmodified records in a raw dataset and create a separate cleaned dataset for analysis.
- Record the collection date, URL, page status, number of reviews retrieved, and any retrieval errors in a scraping log.
- Freeze and version the final dataset before model training and evaluation.

**Table 3.4: Data fields collected from Amazon**

| Data field                | Description                                             | Use in the study                                      |
|---------------------------|---------------------------------------------------------|-------------------------------------------------------|
| Product identifier / ASIN | Stable identifier shown for the listing where available | Product grouping, traceability, and duplicate control |
| Product name and brand    | Displayed smartphone model and manufacturer             | Recommendation display and dataset description        |

| <b>Data field</b>           | <b>Description</b>                          | <b>Use in the study</b>                                                          |
|-----------------------------|---------------------------------------------|----------------------------------------------------------------------------------|
| Product configuration       | Storage, colour, or variant where available | Prevents different configurations from being merged incorrectly                  |
| Product URL                 | Source page used for collection             | Audit trail and reproducibility                                                  |
| Listing price at collection | Displayed price where publicly available    | Dataset description and optional budget filtering; not used as a sentiment label |
| Average product rating      | Displayed aggregate rating                  | Rating-based recommendation baseline                                             |
| Review title                | Title supplied with the review              | Supplementary context for ABSA                                                   |
| Review text                 | Main written review content                 | Primary input to preprocessing and ABSA                                          |
| Review star rating          | Rating attached to the individual review    | Dataset description and consistency checks                                       |
| Review date                 | Date shown for the review                   | Recency description and temporal checks                                          |

### 3.8.3 Data Storage and Security

Raw records are stored unchanged in versioned CSV or JSON files and backed up in approved storage. Cleaned data, aspect records, model splits, and predictions are stored separately. Internal keys preserve traceability without retaining reviewer names or profile links.

## 3.9 Data Preparation and Annotation

### 3.9.1 Preprocessing

Preprocessing removes technical noise while preserving language needed for sentiment. Stop words, negations, intensifiers, emojis, and meaningful punctuation are retained where relevant to the transformer model. Core steps include record validation, duplicate screening,

Unicode and whitespace normalization, language filtering, sentence and clause segmentation, aspect-candidate generation, and leakage-controlled dataset splitting. Copied and near-identical reviews are removed before aggregation, and complete reviews remain within one train, validation, or test partition. These controls respond to the preprocessing and imbalance issues identified in e-commerce and mobile-review research (Rasappan et al., 2024; Lestari et al., 2024).

**Table 3.5: Data preparation procedures**

| Procedure                   | Implementation                                                                                                         | Output                                       |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Record validation           | Confirm product identifier, review text, and source URL.                                                               | Traceable raw record                         |
| Exact duplicate removal     | Compare product identifier and normalized review text.                                                                 | Removal of repeated records                  |
| Near-duplicate screening    | Use text similarity and manual checks for highly similar reviews.                                                      | Reduced copied-content inflation             |
| Text normalization          | Repair encoding, remove HTML remnants, standardize whitespace, and preserve meaningful punctuation.                    | Clean contextual text                        |
| Language filtering          | Retain English reviews for the main model.                                                                             | Consistent language scope                    |
| Sentence segmentation       | Split each review into sentences using NLP sentence boundaries.                                                        | Sentence-level records                       |
| Clause segmentation         | Split contrastive structures such as “but”, “although”, and “while” where separate opinions are expressed.             | Single-opinion or simpler multi-aspect units |
| Aspect candidate generation | Match domain terms and syntactic variants to the six aspect categories.                                                | Candidate aspect-clause pairs                |
| Dataset splitting           | Assign complete reviews to training, validation, or test sets so clauses from one review never enter different splits. | Leakage-controlled model partitions          |

### 3.9.2 Aspect and Sentiment Annotation

A stratified subset covering selected smartphone products and all six review-derived features is manually labelled by the researcher for aspect category and positive, negative,

or neutral sentiment. Opposing opinions are separated into clauses or separate aspect records; “mixed” is not treated as a fourth sentiment class.

A written annotation guide defines each feature and sentiment class. To check consistency without recruiting participants, the researcher will re-label a randomly selected portion of the dataset after a time interval and calculate intra-rater agreement. Inconsistent cases will be reviewed before the final labelled data are divided into 70:15:15 training, validation, and test partitions, with all clauses from the same review kept in one partition.

**Table 3.6: Annotation scheme**

| Element            | Allowed value                                                    | Decision rule                                                                                     |
|--------------------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Aspect category    | Camera; Battery; Display; Performance; Design/Build; Price/Value | Assign every detected opinion to the smartphone feature it evaluates.                             |
| Positive sentiment | Positive                                                         | The clause expresses satisfaction, strength, benefit, or favourable value toward the aspect.      |
| Negative sentiment | Negative                                                         | The clause expresses dissatisfaction, weakness, failure, or unfavourable value toward the aspect. |
| Neutral sentiment  | Neutral                                                          | The aspect is mentioned factually without a clear favourable or unfavourable judgement.           |
| Multiple aspects   | Multiple records                                                 | Create one record per aspect; do not force the whole sentence into a single label.                |
| Opposing opinions  | Clause-level records                                             | Separate the opinions where possible; do not use “mixed” as a substitute for segmentation.        |

## 3.10 System Design and Development

### 3.10.1 Aspect Identification

Aspect identification combines a versioned domain lexicon with noun-phrase and dependency patterns. The lexicon covers explicit terms and synonyms for camera, battery, display, performance, design/build, and price/value, while syntactic rules link features to nearby opinions and reduce false matches (Mansilla et al., 2025).

Terms discovered during pilot analysis may be added with documented justification. The fixed aspect scheme supports transparency, although implicit aspects without recognizable cues may remain undetected.

**Table 3.7: Smartphone aspect categories and indicative terms**

| Aspect       | Indicative terms and expressions                                                      | Operational meaning                                                         |
|--------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Camera       | camera, photo, image, selfie, video, lens, portrait, zoom, night mode                 | Image and video quality, camera functions, and shooting experience          |
| Battery      | battery, battery life, charging, charge, charger, power, drain, backup, fast charging | Endurance, charging speed, and power behaviour                              |
| Display      | display, screen, brightness, resolution, refresh rate, AMOLED, touch, colour          | Visual quality, readability, smoothness, and touch experience               |
| Performance  | speed, processor, RAM, lag, gaming, heat, multitasking, responsiveness                | Processing speed, stability, thermal behaviour, and application performance |
| Design/Build | design, look, weight, body, build, size, grip, finish, premium, durability            | Appearance, ergonomics, materials, and physical construction                |
| Price/Value  | price, cost, value, worth, expensive, cheap, budget, deal, money                      | Affordability and value for money expressed by reviewers                    |

### 3.10.2 Aspect-Level Sentiment Classification

A BERT-based model is fine-tuned for three-class aspect sentiment classification using the target aspect and review clause as input. Transformer models are selected for their contextual performance in complex ABSA and mobile-phone review language (Aziz et al., 2024; Jayakody et al., 2024; Singhal et al., 2025).

The model is compared with a TF-IDF linear SVM baseline. Validation macro F1 determines model selection, class weighting is used where necessary, and the finalized model is tested once on the held-out set. Predicted labels, probabilities, and low-confidence cases are retained for analysis.

### 3.10.3 Feature Score Generation

Aspect records are aggregated by smartphone and feature. Let P, N, and U represent the positive, negative, and neutral review mentions for product p and feature a. In accordance with Research Objective 3, the feature performance score is calculated as the percentage of positive mentions:  $\text{Feature Score}(p,a) = [P(p,a) / (P(p,a) + N(p,a) + U(p,a))] \times 100$ . The score ranges from 0 to 100, where a higher value indicates a larger proportion of positive review evidence for that feature.

Positive, negative, and neutral mention counts are retained and displayed with each score so that the percentage is not presented without its evidence base. At least 10 valid review mentions are required for a smartphone-feature score; products lacking sufficient evidence for a weighted feature are excluded from that recommendation request, and the threshold is examined through sensitivity analysis.

### 3.10.4 User Preference Weighting and Recommendation Ranking

Users assign importance values to the six features. The system normalizes them so that they sum to one:  $\text{Normalized Weight}(a) = \text{Entered Weight}(a) / \sum \text{Entered Weights}$ . If all entries are zero, equal weights are applied.

For each eligible product,  $\text{Final Recommendation Score}(p) = \sum [\text{Normalized Weight}(a) \times \text{Feature Score}(p,a)]$ . Smartphones are sorted from highest to lowest score. This explicit content-based design uses visible review evidence and stated preferences rather than hidden latent profiles (Musto et al., 2017; Liu et al., 2021; Haghighi & Moradian Zadeh, 2025).

**Table 3.8: Example of weighted recommendation scoring**

| Feature     | Feature score | Normalized weight | user | Weighted contribution |
|-------------|---------------|-------------------|------|-----------------------|
| Camera      | 84.0          | 0.30              |      | 25.20                 |
| Battery     | 78.0          | 0.25              |      | 19.50                 |
| Display     | 82.0          | 0.15              |      | 12.30                 |
| Performance | 75.0          | 0.15              |      | 11.25                 |

| Feature                    | Feature score | Normalized user weight | Weighted contribution |
|----------------------------|---------------|------------------------|-----------------------|
| Design/Build               | 70.0          | 0.05                   | 3.50                  |
| Price/Value                | 80.0          | 0.10                   | 8.00                  |
| Final recommendation score | -             | 1.00                   | 79.75                 |

### 3.10.5 Explanation Generation

Explanations are rule-based to prevent unsupported statements. For each recommendation, the module reports the strongest weighted contributions, relevant scores and mention counts, and any important weak feature. The result communicates strengths and trade-offs using evidence derived from the same calculation that produced the rank (Mansilla et al., 2025; Dietz et al., 2022; Danish et al., 2026; Zhu, 2025).

## 3.11 Operationalization of the Framework

The computational framework uses review-derived feature scores and user feature weights as principal inputs, weighted contributions as intermediate values, and recommendation score and rank as outputs. These labels describe algorithmic roles rather than causal social-science constructs.

| A. OPERATIONALIZATION OF IV1       |                                                                                         |                                                                  |             |
|------------------------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------|-------------|
| Variable                           | Parameters / Keywords                                                                   | Operational Measurement                                          | Scale       |
| Camera Sentiment Score             | camera, photo, picture, image, selfie, video, lens, low light, portrait, zoom           | Positive camera mentions / total camera mentions × 100           | Ratio 0–100 |
| Battery Sentiment Score            | battery life, endurance, charging, charge, drain, backup, standby, screen time          | Positive battery mentions / total battery mentions × 100         | Ratio 0–100 |
| Display Sentiment Score            | display, screen, brightness, resolution, color, refresh rate, touch, outdoor visibility | Positive display mentions / total display mentions × 100         | Ratio 0–100 |
| Device Performance Sentiment Score | performance, speed, processor, lag, gaming, multitasking, app loading, heating          | Positive performance mentions / total performance mentions × 100 | Ratio 0–100 |
| Design Sentiment Score             | design, build, look, materials, weight, thickness, ergonomics, durability               | Positive design mentions / total design mentions × 100           | Ratio 0–100 |
| Price & Value Sentiment Score      | price, cost, expensive, affordable, value for money, worth, discount, overprice         | Positive price mentions / total price mentions × 100             | Ratio 0–100 |

| B. OPERATIONALIZATION OF IV2     |                                        |             |
|----------------------------------|----------------------------------------|-------------|
| Variable                         | Operational Measurement                | Scale       |
| Camera Weight                    | User assigns importance to camera      | Ratio 0–100 |
| Battery Weight                   | User assigns importance to battery     | Ratio 0–100 |
| Display Weight                   | User assigns importance to display     | Ratio 0–100 |
| Performance Weight               | User assigns importance to performance | Ratio 0–100 |
| Design Weight                    | User assigns importance to design      | Ratio 0–100 |
| Price & Value Weight             | User assigns importance to price/value | Ratio 0–100 |
| All six weights must total 100%. |                                        |             |

| D. OPERATIONALIZATION OF DEPENDENT VARIABLES |                                                                                                                                                                                                                                          |                                                                  |          |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------|
| Variable                                     | Indicators (Examples)                                                                                                                                                                                                                    | Measurement                                                      | Scale    |
| Recommendation Relevance                     | <ul style="list-style-type: none"> <li>Matches user priorities</li> <li>Recommended phone is suitable</li> <li>Ranking is useful</li> <li>Supports purchase decision</li> <li>Would choose from the recommended list</li> </ul>          | 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree) | Interval |
| Perceived Explainability                     | <ul style="list-style-type: none"> <li>Explanation is clear</li> <li>Easy to understand</li> <li>Feature evidence is visible</li> <li>User understands why one phone ranks higher</li> <li>Transparent reasoning</li> </ul>              | 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree) | Interval |
| User Trust & Acceptance                      | <ul style="list-style-type: none"> <li>Confident in the recommendation</li> <li>Trusts the system's reasoning</li> <li>Satisfied with the results</li> <li>Willing to follow the recommendation</li> <li>Intends to use again</li> </ul> | 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree) | Interval |

| C. OPERATIONALIZATION OF COMPUTATIONAL / INTERVENING VARIABLE |                                                                             |             |
|---------------------------------------------------------------|-----------------------------------------------------------------------------|-------------|
| Variable                                                      | Operational Measurement                                                     | Scale       |
| Weighted Recommendation Score ( $RS_{up}$ )                   | $RS_{up} = \sum_{f=1}^6 (W_{uf} \times FS_{pf}), \quad f = 1 \text{ to } 6$ | Ratio 0–100 |

Note:  $FS_{pf}$  = feature sentiment score of phone  $p$  for feature  $f$ ;  $W_{uf}$  = user weight for feature  $f$ ; positive, negative, and neutral mentions are obtained from ABSA model predictions.

Figure 3.2: Operationalization map of the proposed smartphone recommendation system

Table 3.9: Operationalization of the main variables and outputs

| Variable / output | Role in framework     | Operational definition                                  | Indicator or calculation                                         | Data type / source                        |
|-------------------|-----------------------|---------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------|
| Review text       | Raw evidence          | Written customer opinion linked to a smartphone listing | Review title and review body                                     | Text; scraped Amazon data                 |
| Aspect category   | Intermediate variable | Smartphone feature discussed in an aspect-level clause  | Camera, battery, display, performance, design/build, price/value | Categorical; aspect-identification module |
| Aspect sentiment  | Intermediate variable | Polarity expressed toward the detected aspect           | Positive, negative, or neutral                                   | Categorical; BERT-based classifier        |

| <b>Variable / output</b>   | <b>Role in framework</b>      | <b>Operational definition</b>                                          | <b>Indicator or calculation</b>                                       | <b>Data type / source</b>               |
|----------------------------|-------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------|
| Feature mention count      | Evidence-strength indicator   | Number of valid aspect records for a product-feature pair              | $P + N + U$                                                           | Count; aggregated ABSA output           |
| Feature sentiment score    | Independent/input variable 1  | Percentage of positive review mentions for each smartphone feature     | $[P / (P + N + U)] \times 100$                                        | Continuous 0–100; scoring module        |
| User feature weight        | Independent/input variable 2  | Relative importance assigned by the user to each feature               | Entered weight divided by total entered weights                       | Continuous 0–1; user interface          |
| Weighted contribution      | Mediating computational value | Contribution of one feature to the product score                       | Feature score $\times$ normalized weight                              | Continuous; recommendation engine       |
| Final recommendation score | Dependent/output variable     | Overall fit of a smartphone to the user's priorities                   | Sum of six weighted contributions                                     | Continuous 0–100; recommendation engine |
| Recommendation rank        | Dependent/output variable     | Position of the smartphone after sorting final scores                  | Descending order of final score                                       | Ordinal; recommendation engine          |
| Explanation output         | User-facing output            | Plain-language statement of strengths, trade-offs, and evidence counts | Top contributions, important weakness, feature scores, mention counts | Text; rule-based explanation module     |

| <b>Variable / output</b> | <b>Role in framework</b> | <b>Operational definition</b>              | <b>Indicator or calculation</b> | <b>Data type / source</b>           |
|--------------------------|--------------------------|--------------------------------------------|---------------------------------|-------------------------------------|
| Baseline rating score    | Comparison variable      | Traditional product-level aggregate rating | Displayed average star rating   | Continuous; Amazon product metadata |

## 3.12 Data Analysis and Evaluation

### 3.12.1 Descriptive Analysis

Descriptive analysis reports products, reviews collected and retained, aspect clauses, review dates, rating distribution, available credibility indicators, and aspect and sentiment frequencies. This establishes the composition and coverage of the evidence before model results are interpreted.

### 3.12.2 Aspect Identification Evaluation

Aspect detection is compared with the manually labelled test set using precision, recall, and F1-score for each feature and as macro averages. Missed implicit aspects and incorrect feature assignments are examined through error analysis.

### 3.12.3 Sentiment Classification Evaluation

The BERT classifier and TF-IDF-SVM baseline are evaluated on the same held-out test set. Macro F1 is the primary measure, supported by accuracy, macro precision and recall, class-specific F1, a confusion matrix, and end-to-end aspect-sentiment pair F1.

**Table 3.10: Evaluation metrics used in the study**

| <b>Component</b>      | <b>Metric actually used</b>           | <b>Purpose</b>                                                                                                                           |
|-----------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Aspect identification | Precision, Recall, and Macro F1-score | Evaluates whether the system correctly identifies the six smartphone features: camera, battery, display, performance, design, and price. |

| <b>Component</b>         | <b>Metric actually used</b>                               | <b>Purpose</b>                                                                                                                                                                                                |
|--------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sentiment classification | Accuracy and Macro F1-score                               | Evaluates whether each detected feature is correctly classified as positive, negative, or neutral. Macro F1-score provides balanced evaluation when the sentiment classes contain unequal numbers of records. |
| Sentiment classification | Confusion Matrix                                          | Shows the number of correct and incorrect classifications and identifies common confusion between positive, negative, and neutral sentiment.                                                                  |
| End-to-end ABSA          | Aspect–Sentiment Pair F1-score                            | Evaluates whether the system identifies both the correct smartphone feature and the correct sentiment for the same review clause.                                                                             |
| Recommendation ranking   | Spearman Correlation Rank                                 | Measures the similarity between rankings generated from the development review set and rankings generated from the held-out review set.                                                                       |
| Recommendation ranking   | Top-3 Overlap                                             | Measures how many smartphones appear in the top three recommendations in both the development-based ranking and the held-out review ranking.                                                                  |
| Robustness               | Rank stability under different minimum mention thresholds | Checks whether the smartphone rankings remain reasonably stable when the minimum number of feature mentions is changed.                                                                                       |
| Explanation output       | Faithfulness and Numerical Consistency                    | Checks whether the generated explanation correctly reports the feature scores, feature weights, weighted contributions, review counts, strengths, and weaknesses used to produce the recommendation.          |

| Component          | Metric actually used | Purpose                                                                                                                                                                            |
|--------------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Explanation output | Explanation Coverage | Measures the percentage of recommendations that include all required explanation elements, such as the main contributing features, feature scores, and supporting review evidence. |

### 3.12.4 Recommendation Evaluation

Recommendation evaluation is conducted entirely offline. Review records for each eligible smartphone are divided at review level into development and held-out evaluation subsets. Feature scores and rankings are first calculated from the development reviews for balanced, camera-focused, battery-focused, display-focused, performance-focused, design-focused, and price-focused weight profiles. The same profiles are then applied to feature scores calculated from held-out reviews, which provide an independent reference ranking. Agreement is measured using Spearman rank correlation, NDCG@3, and top-three overlap. The proposed ranking is also compared with an average-rating baseline; better agreement with the held-out feature ranking indicates stronger stability and alignment with unseen review evidence.

### 3.12.5 Explanation Transparency and Faithfulness Evaluation

Explanation evaluation is automated and does not involve participants. For each recommended smartphone, the explanation must identify the feature contributions that actually had the largest effect on the final score, report the correct feature scores and review-mention counts, and disclose an important weak feature when applicable.

Faithfulness is measured as the percentage of explanations whose stated reasons match the computed top weighted contributions. Coverage is the percentage of recommendations for which an explanation is generated. Numerical consistency checks compare every displayed score, weight, contribution, and mention count with the stored calculation. The proposed output is also structurally compared with the average-rating baseline to determine whether feature-level evidence and traceable reasons are available.

## 3.13 Reliability, Validity, and Quality Assurance

### 3.13.1 Dataset Reliability

Reliability is supported through source logs, version control, duplicate checks, separate raw and cleaned datasets, and frozen final data and model partitions.

### 3.13.2 Annotation Reliability

Annotation reliability is supported through a written coding guide, delayed re-annotation of a random subset by the researcher, intra-rater agreement measurement, and correction of inconsistent cases.

### 3.13.3 Model and Internal Validity

Data leakage is controlled by splitting complete reviews, and the held-out test set is protected from tuning. Baseline comparison and error analysis support credible interpretation of performance.

### 3.13.4 Construct and Recommendation Validity

Construct validity is supported by explicit definitions of the six review-derived features and transparent scoring rules. Recommendation validity is assessed through exact verification of weighted-score calculations and agreement between development-based rankings and rankings produced from held-out review evidence.

### 3.13.5 Robustness Checks

Robustness is examined by varying minimum mention thresholds, predefined feature-weight profiles, and review-data splits, while also testing duplicate-sensitive cases and reviewing low-confidence predictions. Stable rankings and explanations under reasonable settings strengthen confidence in the framework.

**Table 3.11: Quality assurance measures**

| Quality area        | Control measure                                        | Expected benefit                                  |
|---------------------|--------------------------------------------------------|---------------------------------------------------|
| Source traceability | Store ASIN, URL, scrape timestamp, and collection log. | Supports auditability and dataset reconstruction. |

| Quality area               | Control measure                                                        | Expected benefit                                                                |
|----------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Raw-data preservation      | Keep an immutable raw dataset separate from all cleaned versions.      | Prevents undocumented alteration of source records.                             |
| Duplicate and spam control | Exact/near-duplicate screening and suspicious-pattern checks.          | Reduces artificially inflated sentiment evidence.                               |
| Annotation consistency     | Written guide, delayed re-annotation, and intra-rater agreement.       | Improves consistency of the labelled benchmark without recruiting participants. |
| Data leakage control       | Keep all clauses from one review in the same split.                    | Prevents overly optimistic test performance.                                    |
| Model comparison           | Evaluate BERT and TF-IDF-SVM on the same held-out data.                | Provides an interpretable performance baseline.                                 |
| Scoring transparency       | Publish equations, mention counts, thresholds, and normalized weights. | Makes recommendation logic auditable.                                           |
| Explanation faithfulness   | Generate text only from computed feature evidence.                     | Prevents unsupported or hallucinated reasons.                                   |
| Robustness analysis        | Test multiple user profiles and evidence thresholds.                   | Shows whether conclusions depend on one configuration.                          |

### 3.14 Ethical Considerations

The study uses only publicly visible review content required for academic analysis. No private account, login-restricted, or payment-protected data are collected, and reviewer names, profile links, or identifiers are excluded from the research dataset. Automated collection will respect applicable platform rules and technical restrictions.

Because no human participants are recruited, informed-consent, questionnaire, interview, and participant-data procedures are not applicable. The study minimizes data collection, retains no reviewer names or profile links, stores only fields required for analysis, and reports results in aggregate. The prototype is a research decision-support tool, not a commercial endorsement or guarantee of product performance.

### 3.15 Tools and Technologies

Table 3.12 lists the principal tools used for collection, preprocessing, modelling, analysis, recommendation, visualization, and version control.

**Table 3.12: Tools and technologies used in the study**

| Component                         | Tool or technology                            | Purpose                                                                                         |
|-----------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------|
| Programming language              | Python                                        | Implements data collection, preprocessing, modelling, scoring, evaluation, and interface logic. |
| Browser-based collection          | Playwright or Selenium                        | Loads permitted dynamic review pages and controls pagination.                                   |
| HTML parsing                      | BeautifulSoup                                 | Extracts data from static page content where applicable.                                        |
| Data handling                     | Pandas, NumPy, CSV/JSON, SQLite or PostgreSQL | Stores, cleans, joins, and aggregates product and review records.                               |
| NLP processing                    | spaCy and regular expressions                 | Sentence/clause segmentation, token processing, noun phrases, and rule-based aspect matching.   |
| Transformer modelling             | Hugging Face Transformers and PyTorch         | Fine-tunes and runs the BERT-based aspect-sentiment classifier.                                 |
| Baseline modelling and evaluation | Scikit-learn                                  | Implements TF-IDF-SVM, classification metrics, and confusion matrices.                          |
| Recommendation engine             | Python scoring functions                      | Normalizes user weights, calculates product scores, and ranks eligible smartphones.             |
| Prototype interface               | Streamlit                                     | Collects feature weights and displays ranked products, evidence, and explanations.              |
| Visualization                     | Matplotlib                                    | Presents dataset distributions, model results, feature profiles, and ranking comparisons.       |
| Version control                   | Git and dated dataset versions                | Tracks source code, lexicon changes, configurations, and reproducible experiment states.        |

## 3.16 Chapter Summary

This chapter presented a pragmatic, predominantly quantitative design-and-development methodology for constructing and evaluating the proposed recommendation system. The process uses a fixed Amazon review dataset, criterion-based product and review selection, controlled preprocessing, manual annotation, hybrid aspect detection, and BERT-based sentiment classification.

Feature scores and normalized user weights generate transparent rankings, while rule-based explanations report strengths, trade-offs, and supporting review evidence. Technical metrics, held-out ranking evaluation, average-rating baseline comparison, automated explanation checks, reliability controls, robustness checks, and ethical safeguards provide the basis for the implementation and results chapters.

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